

Gnana Karthik Yalamanchi

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PROFESSIONAL SUMMARY

Data Scientist with 3+ years of combined professional, internship, and research experience building predictive models, statistical analyses, and machine learning pipelines across enterprise, fintech, and healthcare domains. Skilled in developing classification and behavioral pattern-detection models using Python (Scikit-learn, XGBoost, PySpark), with rigorous evaluation via cross-validation, ROC-AUC, F1, and A/B testing. Experienced engineering features from raw enterprise data (SAP ECC, S/4HANA, BW) into model-ready datasets, then communicating results through Tableau and Power BI dashboards and SHAP-based interpretability reporting for business stakeholders. Proven track record partnering with clinical and business stakeholders to translate ambiguous problems into measurable modeling outcomes. Authorized to work in the US.

TECHNICAL SKILLS

Machine Learning & Statistics: Scikit-learn, XGBoost, Random Forest, Gradient Boosting, Isolation Forest, K-Means Clustering, Classification & Regression, Feature Engineering, Cross-Validation, GridSearchCV, Hypothesis Testing (Chi-Square, T-Test), A/B Testing, Model Evaluation (ROC-AUC, F1, Precision-Recall), SHAP Interpretability, SMOTE

Deep Learning & NLP: BERT, HuggingFace Transformers, SpaCy, LDA Topic Modeling (Gensim), Sentiment Classification, CNN, LSTM, Transformer-based Models

Data Visualization & BI: Tableau, Power BI, Plotly/Dash, Data Modeling, DAX, Executive/Stakeholder Reporting, Data Storytelling

Time Series & Forecasting: Facebook Prophet, ARIMA, LSTM, Demand Forecasting, Seasonality Modeling, MAPE Evaluation

Data Engineering & Big Data: Azure Databricks, PySpark, Spark SQL, Spark Structured Streaming, Apache Kafka, Delta Lake, Medalion Architecture (Bronze/Silver/Gold), Azure Data Factory, Microsoft Fabric Data Factory, dbt Cloud, Great Expectations

Enterprise Data Integration: SAP ERP (ECC, S/4HANA, BW, Datasphere), Structured & Semi-Structured Data, REST APIs, Data Quality & Governance

Cloud, Infrastructure & MLOps: Azure Data Lake, AWS (EC2, S3, Redshift, SageMaker, Lambda), Snowflake, PostgreSQL, MongoDB, FastAPI, MLflow, Model Serving & Deployment

Languages: Python (Pandas, NumPy, PySpark), SQL, TypeScript, Java

ML & Generative AI: Anthropic Claude API, RAG, Agentic Orchestration (MCP), Model Evaluation, Prompt Evaluation

Dev & Reproducibility: Git, Docker, CI/CD, Claude Code, Modular Codebase Design, Technical Documentation

WORK EXPERIENCE

Data & AI Intern

May 2025 – Feb 2026

Sporting Jax & University of North Florida

Jacksonville, FL

- Partnered with coaching and product stakeholders on a player health-tracking application, translating the broad goal of "monitor player health" into a measurable objective: predicting each player's fatigue/injury-risk score from historical performance data, with clear success metrics defined upfront.
- Collected and consolidated player health and performance data (wearable biometrics, training load, game activity logs) spanning 2M+ records using Python (Pandas, NumPy), then conducted exploratory data analysis and statistical profiling to uncover trends and predictive signals in players' historical data.
- Engineered time-series features (rolling averages, workload ratios, recovery indices, fatigue-trend indicators) from each player's historical training and biometric data using Python (Pandas, NumPy, SciPy), handling missing values and normalizing across athletes to build a model-ready dataset.
- Trained and compared models (Logistic Regression, Random Forest, XGBoost) in Scikit-learn on players' historical data to predict fatigue/injury-risk scores, using k-fold cross-validation and grid search to select the best-performing model and reduce overfitting.
- Evaluated model performance using ROC-AUC, F1, and precision-recall tradeoffs, applying SHAP interpretability analysis to explain which factors (training load, recovery time, prior injuries) most influenced each player's predicted health measure.
- Designed and published Tableau dashboards translating predicted risk scores and health trends into visualizations for coaching staff, enabling proactive training-load adjustments and directly influencing prioritization of three product features in the app.
- Prototyped an LLM-powered agentic assistant using Anthropic Claude API to summarize player health alerts for coaching staff, benchmarking output quality against manual review and estimating a 25% reduction in review time.

Graduate Research Assistant – Data and Machine Learning

Jan 2025 – Apr 2025

University of North Florida | Mayo Clinic Research Collaboration

Jacksonville, FL

- Partnered with Mayo Clinic cardiologists to translate clinical requirements into structured feature definitions and a modeling pipeline design, establishing systematic hyperparameter and configuration tracking so any prior result could be reproduced within one command.
- Built a modular clinical data preprocessing pipeline in Python against those requirements, handling noise filtering, standardization, and feature extraction across large-scale multi-lead ECG time-series data; standardized input representations that improved downstream model convergence speed by 20%.

Data Scientist

May 2021 – Feb 2024

ContactPoint 360

Hyderabad, India

- Built predictive churn and revenue forecast models using Python (Random Forest, Gradient Boosting, Linear Regression) and validated with cross-validation, reducing customer attrition signals by 22% across 10+ finance and insurance client accounts.
- Automated ETL pipelines on AWS EC2 with cron-scheduled Python scripts loading transformed data to S3, enabling near-real-time Power BI dashboards and cutting manual data workload by 28%.
- Constructed 30+ Power BI dashboards using DAX and data modeling best practices, standardizing KPIs across \$10M+ revenue portfolios and accelerating financial budget planning cycles for C-suite stakeholders.
- Applied feature engineering and data preprocessing (NumPy, Pandas) to cleanse transactional datasets across 5+ source systems, eliminating 80% redundancy and improving model training data quality.

- Designed and enforced data governance and validation frameworks to meet regulatory reporting standards across financial and insurance domains, supporting audit-readiness for 10+ cross-functional teams.
- Automated ad-hoc reporting workflows using advanced SQL (window functions, CTEs, stored procedures), saving 15+ hours/week in query turnaround and improving stakeholder satisfaction scores.

FEATURED PROJECTS

Customer Churn Prediction & Retention Engine | *Python, XGBoost, Scikit-learn, SHAP, SQL, Tableau*

- Developed a binary classification model using XGBoost and Logistic Regression on 500K+ telecom subscriber records, achieving AUC-ROC of 0.91 and 85% recall on at-risk customers.
- Applied SHAP (SHapley Additive exPlanations) for model interpretability – identified the top 8 churn drivers and presented findings to simulated stakeholders, resulting in a retention strategy that reduced projected churn by 22%.
- Built an automated feature engineering pipeline (usage patterns, billing anomalies, tenure cohorts) and performed hyperparameter tuning via GridSearchCV, improving F1-score from 0.71 to 0.84.
- Deployed the model via a FastAPI REST endpoint and visualized cohort retention lift in a Tableau executive dashboard.

Real-Time Fraud Detection System | *Python, Spark Streaming, Kafka, Databricks, AWS, Scikit-learn*

- Built a real-time anomaly detection pipeline using Apache Kafka and Spark Structured Streaming on Databricks to process 1M+ financial transactions/hour, flagging fraudulent activity with 92% precision and 88% recall.
- Trained ensemble classification models (Random Forest, Gradient Boosting, Isolation Forest) with class imbalance handling (SMOTE), reducing the false-positive rate by 35% versus a rule-based baseline.
- Integrated automated data quality checks with Great Expectations and Databricks Workflows; deployed to an AWS SageMaker serving endpoint with model latency under 50ms per prediction.
- Built an end-to-end MLOps pipeline – feature store, training, evaluation, and deployment – with MLflow experiment tracking and model versioning.

NLP Sentiment & Topic Modeling – Product Review Intelligence | *Python, BERT, SpaCy, HuggingFace, Scikit-learn, Plotly*

- Fine-tuned a BERT-based transformer model (HuggingFace Transformers) on 200K+ e-commerce product reviews for multi-class sentiment classification, achieving 91% accuracy – outperforming a TF-IDF + SVM baseline by 12 points.
- Applied LDA topic modeling (Gensim) to extract 15 latent product themes from unstructured reviews, enabling product teams to prioritize roadmap decisions backed by customer-voice data.
- Built a Plotly/Dash interactive dashboard visualizing sentiment trends by category and time, adopted by a simulated product analytics team to replace 6 hours of weekly manual review.

Sales Forecasting & Demand Planning Model | *Python, Prophet, ARIMA, Snowflake, Power BI, dbt*

- Designed a time series forecasting system comparing Facebook Prophet, ARIMA, and LSTM models on 3 years of multi-SKU retail sales data; Prophet with custom seasonality achieved the lowest MAPE of 6.2%.
- Built a Snowflake-backed data mart with dbt transformations (incremental models, seeds, tests) consolidating 5 data sources into a single forecasting-ready table, reducing data prep time by 70%.
- Delivered a Power BI executive dashboard with demand forecasts, inventory health, and anomaly flags – enabling proactive reorder decisions and reducing simulated stockout events by 18%.

Patient Readmission Risk Prediction – Healthcare ML | *Python, Scikit-learn, Random Forest, SHAP, SQL, Tableau*

- Built a multi-class classification model using Random Forest and Decision Trees to predict 30-day hospital readmission risk for 100K+ diabetes patients, achieving 89% accuracy and an AUC of 0.87.
- Performed statistical hypothesis testing (chi-square, t-test) for feature selection; applied SHAP values to explain individual predictions, ensuring the model met clinical interpretability standards.
- Constructed patient risk segmentation using K-Means clustering to identify 4 distinct high-risk cohorts, enabling targeted intervention programs that reduced simulated readmission rate by 14%.

EDUCATION

Master of Science in Computer Science

University of North Florida

GPA: 3.98 / 4.0

Jacksonville, FL

Relevant Coursework: Artificial Intelligence, Machine Learning, Natural Language Processing, Data Engineering, Big Data Analytics (Databricks/Spark), Statistical Analysis & A/B Testing, Cloud Computing (AWS), Database Systems